

CENTRAL BOARD OF SECONDARY EDUCATION — PRACTICE EXAMINATION
CLASS IX • SCIENCE (SUBJECT CODE: 086) • ACADEMIC SESSION 2026-27

SET F — RAPID FIRE SPEED TEST (30 MINS)

CHAPTER 03: TISSUES: PLANT & ANIMAL STRUCTURES

TIME ALLOWED: 30 MINUTES MAXIMUM MARKS: 25 MARKS SERIES: CBS-IX-SCI-SET-F-03

Student's Name: _____

Roll No.: _____

Class & Section: Class IX - _____

Date of Exam: _____

Invigilator's Sign: _____

GENERAL INSTRUCTIONS (SET F - RAPID FIRE SPEED TEST):

1. The question paper comprises **25 Rapid Fire Multiple Choice Questions** of **1 mark each**.
2. Designed for **Rapid Recall & High-Speed Accuracy**: Recommended target time is **30 Minutes**.
3. All questions are compulsory. There is no negative marking.
4. Darken the corresponding bubble on the **OMR Sheet** completely using a blue/black ballpoint pen.

 **SPEED CHALLENGE INSTRUCTIONS:** Answer promptly without looking at notes. Test your instant recall of formulas, scientific definitions, and core NCERT concepts!

Q.1 Which meristematic tissue is located at the growing root tips and shoot apex?

- | | |
|--------------------------|----------------------|
| (A) Lateral meristem | (B) Apical meristem |
| (C) Intercalary meristem | (D) Vascular cambium |

Q.2 The increase in girth (diameter) of a tree trunk is due to:

- | | |
|--------------------------|--------------------------------|
| (A) Apical meristem | (B) Lateral meristem (Cambium) |
| (C) Intercalary meristem | (D) Parenchyma |

Q.3 Which plant tissue contains living cells with localized wall thickenings of pectin at cell corners?

- | | |
|------------------|-------------------|
| (A) Sclerenchyma | (B) Collenchyma |
| (C) Parenchyma | (D) Xylem vessels |

Q.4 Coconut husk is made of which dead mechanical tissue?

- | | |
|-------------------------|----------------------|
| (A) Collenchyma | (B) Aerenchyma |
| (C) Sclerenchyma fibres | (D) Xylem parenchyma |

Q.5 Which simple permanent tissue forms the bulk of soft plant organs and stores food?

- | | |
|-----------------|-------------------|
| (A) Parenchyma | (B) Sclerenchyma |
| (C) Collenchyma | (D) Phloem fibers |

Q.6 Aerenchyma tissue provides buoyancy to aquatic plants because it contains:

- | | |
|------------------------|------------------------------|
| (A) Large air cavities | (B) Dense oil droplets |
| (C) Lignin deposits | (D) Calcium oxalate crystals |

Q.7 Which component of phloem tissue conducts organic nutrients (sucrose) through perforated end walls?

- | | |
|-----------------|-------------------|
| (A) Sieve tubes | (B) Phloem fibers |
| (C) Tracheids | (D) Xylem vessels |

Q.8 Which xylem component is the only living element in mature xylem?

- | | |
|------------------|----------------------|
| (A) Tracheids | (B) Xylem vessels |
| (C) Xylem fibres | (D) Xylem parenchyma |
-

Q.9 Which epithelial tissue lines the alveoli of the lungs and capillary blood vessels for rapid diffusion?

- | | |
|--------------------------------|--------------------------|
| (A) Simple Squamous Epithelium | (B) Ciliated Columnar |
| (C) Stratified Cuboidal | (D) Glandular Epithelium |
-

Q.10 Which tissue lines the respiratory windpipe (trachea) to sweep trapped dust and mucus outward?

- | | |
|----------------------------------|-------------------------|
| (A) Ciliated Columnar Epithelium | (B) Squamous Epithelium |
| (C) Hyaline Cartilage | (D) Areolar Tissue |
-

Q.11 Tendons are tough fibrous connective tissues that connect:

- | | |
|---------------------|-----------------------------|
| (A) Bone to bone | (B) Skeletal muscle to bone |
| (C) Nerve to muscle | (D) Cartilage to muscle |
-

Q.12 Ligaments join:

- | | |
|--------------------|-------------------------|
| (A) Muscle to bone | (B) Bone to bone |
| (C) Skin to muscle | (D) Cartilage to tendon |
-

Q.13 Which tissue stores fat beneath the skin and around internal organs, serving as a thermal insulator?

- | | |
|--------------------------|---------------------|
| (A) Adipose tissue | (B) Areolar tissue |
| (C) Dense regular tissue | (D) Squamous tissue |
-

Q.14 The matrix of bone tissue is made exceptionally hard and rigid by deposits of:

- | | |
|----------------------------------|------------------------------------|
| (A) Calcium and Phosphorus salts | (B) Sodium and Potassium chlorides |
| (C) Iron and Magnesium | (D) Cellulose and lignin |
-

Q.15 Cartilage is present at all the following anatomical sites EXCEPT:

- | | |
|--------------------|---------------------|
| (A) Ear pinna | (B) Tip of the nose |
| (C) Tracheal rings | (D) Kidney tubules |
-

Q.16 Which type of muscle tissue exhibits branched fibers and intercalated discs?

- | | |
|---------------------|---------------------|
| (A) Skeletal muscle | (B) Smooth muscle |
| (C) Cardiac muscle | (D) Visceral muscle |
-

Q.17 Smooth (unstriated) muscle tissue is found in:

- | | |
|----------------|---|
| (A) Leg biceps | (B) Wall of the stomach and blood vessels |
| (C) Tongue | (D) Forearm triceps |
-

Q.18 The functional transmitting cell of nervous tissue is the:

- | | |
|---------------|-------------------------|
| (A) Nephron | (B) Neuron (Nerve cell) |
| (C) Neuroglia | (D) Sarcomere |
-

Q.19 The long, slender conducting cytoplasmic projection extending from the neuron's cell body is the:

- | | |
|--------------|------------|
| (A) Dendrite | (B) Axon |
| (C) Synapse | (D) Myelin |
-

Q.20 The junctional gap between two adjacent communicating neurons is called a:

- (A) Centrosome (B) Synapse
(C) Intercalated disc (D) Node of Ranvier

Q.21 Which cellular component of mammalian blood lacks a cell nucleus at maturity?

- (A) White Blood Cell (Leukocyte) (B) Red Blood Cell (Erythrocyte)
(C) Macrophage (D) Osteocyte

Q.22 Which formed blood element is essential for initiating blood clot formation at a wound site?

- (A) Platelets (Thrombocytes) (B) Erythrocytes
(C) Eosinophils (D) Lymphocytes

Q.23 The waxy, waterproof chemical substance deposited on the cork cell walls of tree bark is:

- (A) Cutin (B) Suberin
(C) Pectin (D) Chitin

Q.24 Which tissue acts as a packing and binding filler tissue between skin and underlying muscles?

- (A) Areolar connective tissue (B) Dense regular tissue
(C) Stratified epithelium (D) Hyaline cartilage

Q.25 Desert plants (xerophytes) minimize transpiration primarily by having a thick outer coating of:

- (A) Lignin (B) Cutin (Cuticle)
(C) Suberin (D) Cellulose

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET F)

01 (A) (B) (C) (D)	02 (A) (B) (C) (D)	03 (A) (B) (C) (D)	04 (A) (B) (C) (D)	05 (A) (B) (C) (D)
06 (A) (B) (C) (D)	07 (A) (B) (C) (D)	08 (A) (B) (C) (D)	09 (A) (B) (C) (D)	10 (A) (B) (C) (D)
11 (A) (B) (C) (D)	12 (A) (B) (C) (D)	13 (A) (B) (C) (D)	14 (A) (B) (C) (D)	15 (A) (B) (C) (D)
16 (A) (B) (C) (D)	17 (A) (B) (C) (D)	18 (A) (B) (C) (D)	19 (A) (B) (C) (D)	20 (A) (B) (C) (D)
21 (A) (B) (C) (D)	22 (A) (B) (C) (D)	23 (A) (B) (C) (D)	24 (A) (B) (C) (D)	25 (A) (B) (C) (D)

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & RAPID EXPLANATIONS (SET F)

Q. No.	Correct	Scientific Rationale & Formula Summary
Q.1	(B)	Shoot and root apical meristems are located at terminal tips and drive primary longitudinal growth.
Q.2	(B)	Lateral meristems (vascular and cork cambium) produce radial secondary growth, increasing trunk girth.
Q.3	(B)	Collenchyma cells have corner pectin-cellulose thickenings that provide flexibility and support to growing organs.
Q.4	(C)	Coir from coconut husk is composed of dead sclerenchyma fibers with heavily lignified, cemented walls.
Q.5	(A)	Parenchyma is the most abundant unspecialized ground tissue, performing storage, photosynthesis, and secretion.
Q.6	(A)	Aerenchyma is modified parenchyma with large interconnected intercellular air spaces that help water plants float.
Q.7	(A)	Sieve tube elements connect longitudinally with perforated sieve plates to translocate photosynthesized sugars.
Q.8	(D)	Tracheids, vessels, and fibers are dead at maturity; only xylem parenchyma remains living for lateral storage.
Q.9	(A)	Simple squamous epithelium forms an ultra-thin, smooth sheet of pavement-like cells optimizing gaseous diffusion.
Q.10	(A)	Ciliated columnar cells bear beating hair-like cilia that sweep mucus and trapped particulates up the respiratory tract.
Q.11	(B)	Tendons connect muscle to bone; ligaments connect bone to bone across movable joints.
Q.12	(B)	Ligaments are elastic fibrous cords containing elastin that stabilize joints by binding bones together.
Q.13	(A)	Adipose tissue contains adipocytes packed with fat globules, providing energy storage and thermal insulation.
Q.14	(A)	Bone matrix consists of collagen fibers mineralized with calcium phosphate (hydroxyapatite) crystals.
Q.15	(D)	Kidney tubules are lined by cuboidal epithelium; ear pinna, nose tip, larynx, and trachea contain cartilage.
Q.16	(C)	Cardiac muscle fibers are involuntary, cylindrical, branched, and joined by electrical intercalated discs.
Q.17	(B)	Involuntary unstriated spindle-shaped smooth muscle lines internal visceral organs such as the gut and blood vessels.
Q.18	(B)	Neurons are electrically excitable cells specialized to receive, conduct, and transmit nerve impulses.
Q.19	(B)	The axon conducts electrical action potentials away from the neuronal soma (cell body) toward synaptic terminals.
Q.20	(B)	The synapse is the microscopic gap across which neurotransmitters diffuse to propagate electrical signals between neurons.
Q.21	(B)	Mature mammalian RBCs are enucleated biconcave discs optimized to carry oxygen bound to hemoglobin.
Q.22	(A)	Platelets aggregate at vascular lesions and release thromboplastin to trigger the coagulation cascade.
Q.23	(B)	Suberin is a hydrophobic waxy biopolymer that makes mature dead cork cells impermeable to water and gases.
Q.24	(A)	Areolar tissue is loose connective tissue that fills spaces inside organs, supports internal organs, and helps tissue repair.
Q.25	(B)	A thick waxy cuticle layer of cutin coats the epidermal surface of desert plants to drastically curtail water loss.